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1.3.7 Equivalence of matrix norms

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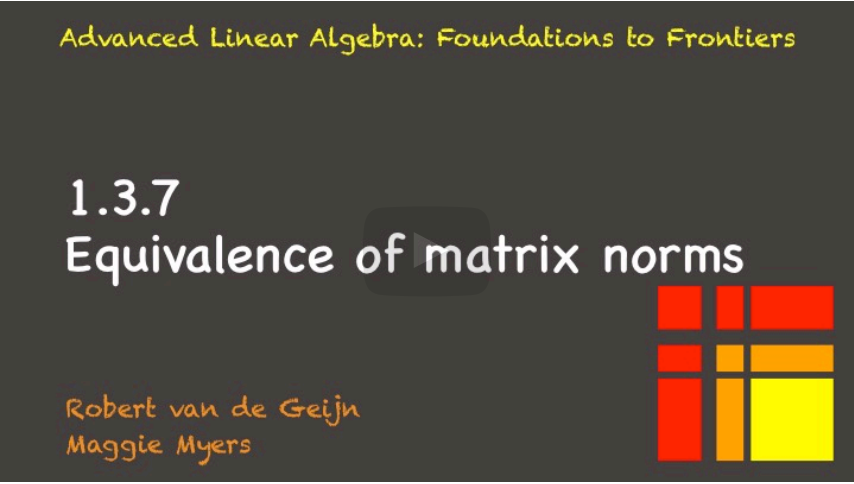
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When we talked about vector norms, we made a big deal out of the fact that if a vector was considered small in one norm, then it was also small in all other norms. And if it was large in one norm, then it was large in all other norms. And we called that equivalence

Video

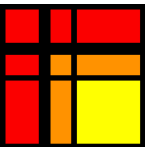
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Reading assignment

1.0/1.0 point (graded)



Read Unit 1.3.7 of the notes. [\[LINK\]](#)

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? why_sqrt(n) vs sqrt(m)?

In the matrix of equivalences below, it says that $\text{norm2}(A) \leq \sqrt{n} \text{norm1}(A)$, where n is the #...

2

Homework 1.3.7.1.

1/1 point (graded)
Fill out the following table:

A	$\ A\ _1$	$\ A\ _\infty$	$\ A\ _F$	$\ A\ $
$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	1	1	1.732	1
	✓ Answer: 1	✓ Answer: 1	✓ Answer: sqrt(3)	✓
$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$	4	3	3.464	3.4
	✓ Answer: 4	✓ Answer: 3	✓	✓
			Answer: sqrt(12)	Answer: sqrt(12)
$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{pmatrix}$	3	1	1.732	1.732
	✓ Answer: 3	✓ Answer: 1	✓ Answer: sqrt(3)	✓

For full solution see notes: [LINK]

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Homework 1.3.7.2

1/1 point (graded)
Given $A \in \mathbb{C}^{m \times n}$ show that $\|A\|_2 \leq \|A\|_F$.

Hmmm, actually, this is really easy to prove once we know about the SVD... Hard to prove without it. So, this problem will be moved...

☒ done



For full solution see notes: [LINK]

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Homework 1.3.7.3

1/1 point (graded)
Let $A \in \mathbb{C}^{m \times n}$. The following table summarizes the equivalence of various matrix norm:

	$\ A\ _1 \leq \sqrt{m}\ A\ _2$	$\ A\ _1 \leq m\ A\ _\infty$	$\ A\ _1 \leq \sqrt{m}\ A\ _F$
$\ A\ _2 \leq \sqrt{n}\ A\ _1$		$\ A\ _2 \leq \sqrt{m}\ A\ _\infty$	$\ A\ _2 \leq \ A\ _F$
$\ A\ _\infty \leq n\ A\ _1$	$\ A\ _\infty \leq \sqrt{n}\ A\ _2$		$\ A\ _\infty \leq \sqrt{m}\ A\ _F$
$\ A\ _F \leq \sqrt{n}\ A\ _1$	$\ A\ _F \leq ?\ A\ _2$	$\ A\ _F \leq \sqrt{m}\ A\ _\infty$	

For each, prove the inequality, including that it is a tight inequality for some nonzero A .

(Skip $\|A\|_F \leq ?\|A\|_2$: we will revisit it in Week 2.)

☒ done



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